

SANKUR KUNDU

Machine Learning Engineer • Software Engineer • Data Scientist

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Engineer who ships systems end to end. Co-founder and Director of Technology at Gigzen, where I built both products alone: Buzz Buzz, a 13K-line Android app on Postgres row-level security, and Populace, a testing platform that surfaces the multi-user bugs one developer cannot reach. All of it public, each README stating what the code measures — including where it misses.

EXPERIENCE

Freelance — Data Scientist — *Contract via Upwork*

Remote, client in Spain | Jun 2026 – Present

- Engineer Python/Pandas/NumPy/SQL pipelines over multi-GB census, commercial and STATA .dta sources, turning multi-year raw data into analysis-ready datasets for an international research team.
- Collaborate cross-functionally with researchers and statisticians, building validation and reproducibility checks into every deliverable under fixed publication timelines.

Gigzen Private Limited — *Co-founder & Director, Technology*

Bhubaneswar, India | Apr 2026 – Present

- Co-founded an MCA-registered company targeting the USD 582B gig economy and lead its technical function, owning the products end to end: architecture, implementation, database design, release engineering and compliance.
- Run the production Postgres both products share — 17 tables under 48 row-level-security policies — and the signed Android release pipeline.

PRODUCTS

Populace — *Multi-user testing platform*

[Shakhtar-Sankur/populace](https://shakhtar-sankur.github.io/populace)

- Designed an agent-based platform where concurrent synthetic users exercise a real app through its own API as real authenticated accounts, surfacing presence, sync, permission and fan-out bugs one developer cannot reach.
- Built an app-agnostic engine with zero runtime dependencies behind a 13-method adapter contract, per-endpoint p50/p95/p99 latency instrumentation that exposes scalability limits before real users do, and three independent guards that refuse to run against production; exits non-zero so it gates CI.
- Made honest reporting a design constraint: an empty adapter method does not count as coverage, so a fresh scaffold reports 2/13 and refuses to run rather than passing while testing nothing. 20/20 self-tests against an in-memory adapter.

Buzz Buzz — *Gig-worker platform, Android*

[Shakhtar-Sankur/buzz-buzz](https://shakhtar-sankur.github.io/buzz-buzz)

- Built a React + TypeScript + Capacitor app on Supabase: 17 Postgres tables under 48 row-level-security policies, realtime sync, presence, read receipts, GPS tracking and groups. Posts and messages written with no signal queue on the device and send on reconnect.
- Localized into 16 languages at full parity — 278 keys each, consent and legal text included — mirrored right-to-left for Arabic, with 27 currencies auto-selected across 49 countries.
- Debugged a recursive row-level-security policy that silently broke every chat read, fixed it with a SECURITY DEFINER helper, and verified end to end against a live backend. Measured photos at 479 KB each as base64 in Postgres — up to 9.4 MB to open a feed — and moved them to object storage with ~20 KB thumbnails.

SELECTED ML PROJECTS

- ML Serving Platform** — The operational layer around inference: versioned rollout, A/B routing, circuit breaking, per-feature drift detection, caching and metrics. *Flask, Redis, Triton, Kubernetes, Prometheus, Docker*
- Edge AI Model Compression** — Compression pipeline for battery-powered edge inference: distillation from a ResNet50 teacher, magnitude pruning, int8 quantization, ONNX export. Targeted 8× smaller and measures -6.3%, with the reason published beside it. *PyTorch, ONNX, ONNX Runtime*
- IoT Predictive Maintenance on GCP** — LSTM failure prediction on a 48-hour forecast horizon with chronological splits, Optuna tuning, FastAPI serving and a GCP deployment path. *TensorFlow, Optuna, FastAPI, Pub/Sub, BigQuery, Vertex AI*
- DETEK — Data-Leak Detection on AWS** — Pairs a BERT content classifier with an LSTM behavioural model, because either alone produces unusable alerts. CloudTrail and VPC Flow collectors, Terraform infrastructure. *PyTorch, Transformers, Flask, boto3, Terraform*

SKILLS

Languages: Python, TypeScript, SQL, R, STATA

ML & AI: Production ML pipelines, model training and evaluation, statistical modeling, deep learning (LSTM, BERT, DDPM), knowledge distillation, pruning, int8 quantization, ONNX export, anomaly detection, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Systems & Cloud: System design, distributed systems, CI/CD, testing, GCP (BigQuery, Vertex AI, Dataflow, Pub/Sub), AWS (Lambda, S3, CloudTrail), FastAPI, Flask, Node.js, React, Capacitor, PostgreSQL and Supabase (row-level security), Redis, Triton, MLflow, Optuna, Prometheus, Docker, Kubernetes, Terraform

EDUCATION & CERTIFICATIONS

Institute of Engineering and Management, Kolkata — B.Tech, Computer Science & Engineering

2022 – 2026

Coursework: Data Structures & Algorithms, Operating Systems, Databases, Computer Networks, Machine Learning

Certifications: Meta Data Analyst; Google Data Analytics; Google Cybersecurity; IBM Cybersecurity Essentials; Google Cloud and AWS engineering paths; Microsoft Azure AI Cloud Apps